Q. Transport system in vertebrates?

- Vertebrates have a closed circulatory system in which the blood never leaves the blood vessels.
- Blood moves from the heart, through arteries, arterioles, capillaries, venules, veins, and back to the heart.
- Exchange between the blood and extracellular fluid only occurs at the capillary level.
- The heart is the center to the circulatory system that circulates the blood throughout the body.
- The coordination of the circulatory system and the respiratory system to ensure proper gas exchange is very important in animals that have lungs and gills.

Q. Define Arteries?

- Arteries are elastic blood vessels that carry blood away from the heart to the organs and tissues of the body.
- The central canal of an artery (and of all blood vessels) is a lumen.

Q. Define veins?

- Veins are relatively inelastic and the wall of a vein contains the same three layers (tunicae) as arterial walls, but the middle layer is much thinner.
- One or more valves are present which permit blood flow only in one direction.

Q. Define capillaries?

- Capillaries are generally composed of a single layer of endothelial cells and are the most numerous blood vessels in an animal's body.
- Capillaries make an enormous surface area available for the exchange of gases, fluids, nutrients, and wastes between the blood and nearby cells.

Q. What is First line of defense?

First line of defense is external. It includes

- Skin

- Mucous membranes and their secretions.

Q. What is Second line of defense?

Second line of defense is internal. It includes

- Phagocytic WBCs
- Antimicrobial proteins
- Inflammatory response

Q. Write the process of Exhalation in humans?

The out flow of air is called Exhalation. It is also called as Expiration.

- The intercostal muscles and the diaphragm relax, allowing the thoracic cavity to return to its original, smaller size and increasing the pressure in the thoracic cavity.
- Abdominal muscles contract, pushing the abdominal organs against the diaphragm, further increasing the pressure within the thoracic cavity.
- The action in step 2 causes the elastic lungs to contract and compress the air in the alveoli.
- With this compression, alveolar pressure becomes greater than atmospheric pressure, causing air to be expelled (exhaled) from the lungs.

Q. What is Feedback system?

Feedback system is the system in which the controlling mechanism is itself controlled by the product of the reaction it is controlling.

Example: Incubator, Toaster, Iron.

- Positive Feedback
- Negative Feedback

Q. Write the role of chromaffin tisues?

- Chromaffin tissue is a group of neuroendocrine cells in the adrenal medulla that synthesizes steroid hormones.

- These hormones are adrenaline (epinephrine) and noradrenaline (norepinephrine)
- These hormones prepare some vertebrates for stressful emergency situations (fight and flight)

Q. Define external Respiration?

The process of acquiring oxygen and eliminating carbon dioxide is called external respiration. The exchange of oxygen and carbon dioxide occurs across respiratory surfaces.

Q. Write the process of Inhalation in humans?

The intake of air is called as Inhalation. It is also called as inspiration.

- Several sets of muscles, the main ones being the diaphragm and intercostal muscles, contract.
- The intercostal muscles stretch from rib to rib, and when they contract, they pull the ribs closer together, enlarging the thoracic cavity.
- The thoracic cavity further enlarges when the diaphragm contracts and flattens.
- The increased size of the thoracic cavity causes pressure in the cavity to drop below the atmospheric pressure.
- Air rushes into the lungs, and the lungs inflate.

Q. What is Hemocyanin?

- Hemocyanin is the most commonly occurring respiratory pigment in molluscs and certain crustaceans.
- Hemocyanin contains metallic copper, has a bluish color when oxygenated, and always occurs dissolved in hemolymph.

Q. What is serum?

Serum is plasma from which fibrinogens have been removed.

Q. Difference between Lymph and Blood?

- Lymph is colorless fluid where as Blood is colored fluid.
- Lymph is the part of lymphatic system whereas blood is the part of Blood circulatory system.
- Lymph contains plasma and a lesser number WBCs and platelets, whereas blood contains plasma, RBCs, WBCs and Platelets. • Lymph contains low amount of O2 whereas blood carries large amount of O2.
- Lymph lacks protein, whereas blood plasma has Ca+2 & PO4 -3.
- Flow of lymph is slow where flow of blood is fast.

Q. What is chemical messenger?

Chemical messengers are involved in

- Communication.
- Maintenance of homeostasis in an animal's body.
- Body's response to various stimuli.

Q. What is neuropeptides?

- Neuropeptides are specialized neurons (called neurosecretory cells) which secrete neuropeptides (neurohormones).
- The blood or other body fluids transport neuropeptides to nonadjacent target cells, where neuropeptides exert their effects.

Q. What is lymphatic system?

The lymphatic system is part of the vascular system.

- The system begins with network of lymphatic vessels that carry a clear fluid called lymph.
- The small lymphatic capillaries merge to form larger lymphatic vessels called lymphatics.
- Lymphatics are thin-walled vessels with valves that ensure the one-way flow of lymph.

- Lymph is the extracellular fluid that seeps from the bloodstream and accumulates in the lymph vessels.
- The lymphatic vessels pass through the lymph nodes on their way back to the heart.

Q. What is vaccine?

Vaccine is a substance used to stimulate the production of antibodies and provide immunity against one or several diseases. It prepared from the causative agent of a disease or its products.

Q. What is fibrinogen?

Fibrinogen is a protein that is necessary for blood coagulation (clotting)

Q. Write the pumping action of Human heart?

- Adult human heart beats or pumps 72 times a minute.
- In the beating of the heart the atria contract simultaneously and ventricles immediately after words.
- The deoxygenated blood from the lower and upper part of the body is brought into the right atrium.
- On the contraction of the ventricles the blood from the right ventricle is taken to the lungs by the pulmonary artery.
- The valves open and close due to blood pressure changes when the heart contracts during each heartbeat.

Q. What are exocrine glands?

Exocrine Glands are those glands which secrete chemicals into ducts that, in turn, empty into body cavities or onto body surfaces (e.g., mammary, salivary, and sweat glands).

Q. What are endocrine glands?

Endocrine Glands are those glands, have no ducts and secrete chemical messengers, the hormones, directly into the tissue space next to each endocrine cell.

Q. Where are the kidney present And how do they functions?

Q. Write the Functions of kidney Harmones?

Kidneys are located on the both side of the spine at the lowest level of the rib cage.

Kidney produces three hormones which performs different functions.

- **Erythropoietin:** It stimulates red blood cell production, the target is bone marrow.
- **Urotensin:** It stimulates constriction of the arteries.
- **Calcitrol:** It helps in the absorption of dietary calcium and phosphorus from small intestine.

Q. Write the name of major endocrine glands?

The major endocrine glands are

- Hypothalamus
- Pituitary
- Thyroid,
- Parathyroids
- Adrenals
- Pineal
- Thymus

Q. Hepatic Portal System?

- In hepatic portal system nutrients from the small intestine and the right half of the colonare absorbed into blood through the bed of capillaries.

- This blood then moves to the liver through the Hepatic Portal Vein to store and metabolize in liver.
- From the liver the blood enters main blood stream via hepatic vein into the inferior vena cava to the heart.

Q. Write the name of different types of vaccines?

Different types of vaccines are

- Live-attenuated vaccines
- Inactivated vaccines
- Toxoid vaccines
- Attenuated or Weakened vaccines

Q. How arthropods form its new cuticle?

The Corpora Cardiaca produces thoracotropic hormone, which is carried to the prothoracic glands, stimulating them to produce and release ecdysone, which induces molting, the reabsorption of some of the old cuticle and the development of a new cuticle.

Q. Write the Heart beat rate in humans?

Adult human heart beats or pumps 72 times a minute

Q. What is Active immunity?

The immunity which results from the production of antibodies by the immune system in response to the presence of an antigen is called Active immunity. It comes from exposure to a pathogen.

Q. What is Passive immunity?

Passive immunity is conferred from outside the body, so it doesn't require exposure to an infectious agent or its antigen.

Q. Define Antigens?

The foreign substance that elicits the immune response is called an antigen.

Q. Define Antibodies?

The immune system responds to an antigen by producing specific proteins called antibodies.

Q. What is diaphragm?

Large sheet of muscle present below the lungs that controls their breathing is called diaphragm. It is the principal muscle of respiration.

Q. What is tympanum?

Ans. Tympanum is modified integument stretched oner cartilaginous ring. It is also called the eardrum.

Q. What is lymphatic system?

Q. What is vertebrate lymphatic system?

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- The system begins with network of lymphatic vessels that carry a clear fluid called lymph.
- The small lymphatic capillaries merge to form larger lymphatic vessels called lymphatics.
- The lymphatic vessels pass through the lymph nodes on their way back to the heart.

Q. Which major class of hormones produced in ovaries?

Major classes of hormones produced in ovaries are

- Estrogens (Estrin , Estrone , and Estradiol)
- Progestins
- Relaxin
- Inhibin

Q. Write down the components of Transport System?

The components of Transport System are

- Circulatory Fluid (blood)
- Pumping device (heart)
- Vessels (arteries, veins, capillaries)

Q. What is Bimodal and Trimodel Breathing?

- **Bimodal breathing** is the ability of an organism to exchange respiratory gases simultaneously with both air and water.
- **Trimodal breathing** is the ability of an organism to exchange respiratory gases through three surfaces i.e. skin, gills, and lungs

Q. Write the name of respiratory pigments?

The names of respiratory pigments are

- Hemoglobin
- Hemocyanin
- Hemerythrin
- Chlorocruorin

Q. What does it mean when your lymphocytes are high.

When your lymphocytes are high it indicates your body is dealing with an infection or other inflammatory condition.

Q. What is cutaneous respiration?

Ans. Cutaneous Respiration is the exchange of gases through skin.

Q. What is Thyroid gland?

- The thyroid gland is located in the neck, anterior to the trachea.
- Two of its secretions are thyroxine and triiodothyronine, both of which influence overall growth, development, and metabolic rates.
- Another thyroid hormone, calcitonin, helps control extracellular levels of calcium ions.

Q. Thymus gland?

- The thymus gland is a small organ behind the breastbone (sternum) that plays an important function both in the immune system and endocrine system.
- It is only active until puberty.
- After puberty, the thymus starts to shrink slowly and become replaced by fat.
- Thymus produces the hormone thymosin.

Q. Write down three main types of lymphocytes?

Three main types of lymphocytes are

- T cells
- B cells
- Natural killer cells

Best of luck

Thanks

@MWK@